Anexo A - GM-3 - UML completo de DEADBEAT

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Inventario anterior conservado para las clases sin cambios. GameWorld actualizado y contrastado con su arbol sintactico Java final; no se modificaron las otras 28 fuentes de produccion respecto del inventario anterior. El documento separa la vista conectada de relaciones y los compartimentos completos de cada clase para mantener legibilidad.

Cobertura. 29 archivos de produccion y 38 tipos propios de produccion. Incluye clases, interfaces, enums, tipos anidados y la clase anonima del cache de CityMap. Se conservan tambien siete clases e interfaces de apoyo de libGDX.

Miembros propios de produccion. 344 atributos, 27 constructores explicitos y 281 metodos declarados. Diferencias Java/UML: cero. Los miembros de libGDX se contabilizan por separado: 1 atributo, 0 constructores explicitos y 42 metodos declarados.

Organizacion. La vista conectada siguiente conserva las relaciones de produccion. La trazabilidad Java/UML y las fichas de este anexo incluyen todos los miembros declarados de los 38 tipos propios de produccion y de los siete tipos de apoyo de libGDX.

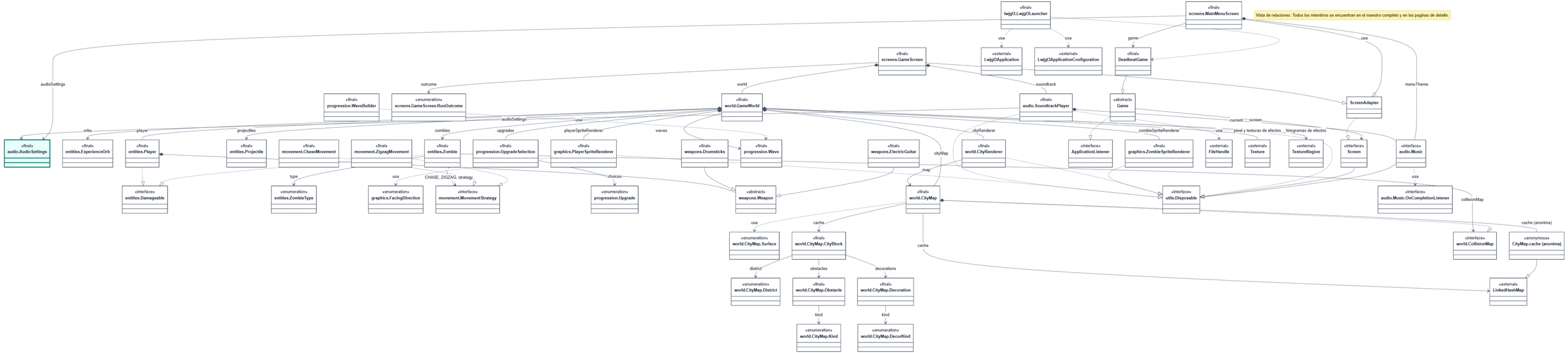
Lectura. + publico; - privado; # protegido; ~ acceso de paquete. Miembros static subrayados, metodos abstractos en cursiva. Las restricciones se escriben entre llaves; no se copian miembros heredados ni se inventan constructores implicitos.

Relaciones. Triangulo continuo: herencia; triangulo discontinuo: realizacion; flecha continua: asociacion; flecha discontinua: dependencia; rombo lleno: composicion. Las composiciones se basan en creacion y propiedad del objeto, no solo en final.

Alcance externo. Se incluyen completas las pequenas bases de libGDX que enlazan las pantallas, el juego y Music. Texture, TextureRegion, FileHandle, las clases externas del launcher y LinkedHashMap son referencias externas de frontera. No se expande recursivamente toda la biblioteca.

Advertencia de formato. El maestro PNG es un poster de consulta con zoom. Para Google Docs se incluyen imagenes individuales de cada clase y secciones. Las paginas de miembros de este PDF son A4 y mantienen texto legible, continuando las clases extensas en la pagina siguiente.

Vista conectada de relaciones



Trazabilidad Java / UML

A: atributos. C: constructores explicitos. M: metodos. Cada constante enum cuenta como atributo. Cada sobrecarga cuenta por separado. Los valores J/U comparan Java con los compartimentos completos de este anexo. GameWorld: 27 atributos, 3 constructores y 34 metodos.

Clase / interfaz	A J/U	C J/U	M J/U	Dif.
audio.AudioSettings	3 / 3	1 / 1	7 / 7	0
audio.SoundtrackPlayer	16 / 16	2 / 2	22 / 22	0
DeadbeatGame	0 / 0	0 / 0	2 / 2	0
entities.Damageable	0 / 0	0 / 0	2 / 2	0
entities.ExperienceOrb	3 / 3	1 / 1	3 / 3	0
entities.Player	24 / 24	1 / 1	29 / 29	0
entities.Projectile	6 / 6	1 / 1	7 / 7	0
entities.Zombie	17 / 17	2 / 2	16 / 16	0
entities.ZombieType	8 / 8	1 / 1	0 / 0	0
graphics.FacingDirection	9 / 9	0 / 0	1 / 1	0
graphics.PlayerSpriteRenderer	6 / 6	1 / 1	2 / 2	0
graphics.ZombieSpriteRenderer	19 / 19	1 / 1	10 / 10	0
movement.ChaseMovement	0 / 0	0 / 0	1 / 1	0
movement.MovementStrategy	0 / 0	0 / 0	1 / 1	0
movement.ZigzagMovement	0 / 0	0 / 0	1 / 1	0
progression.Upgrade	7 / 7	1 / 1	2 / 2	0
progression.UpgradeSelection	4 / 4	0 / 0	7 / 7	0
progression.Wave	4 / 4	1 / 1	3 / 3	0
progression.WaveBuilder	4 / 4	0 / 0	6 / 6	0
screens.GameScreen	70 / 70	2 / 2	36 / 36	0
screens.GameScreen.RunOutcome	3 / 3	0 / 0	0 / 0	0
screens.MainMenuScreen	30 / 30	1 / 1	17 / 17	0
weapons.Drumsticks	0 / 0	1 / 1	1 / 1	0
weapons.ElectricGuitar	0 / 0	1 / 1	1 / 1	0
weapons.Weapon	3 / 3	1 / 1	6 / 6	0
world.CityMap	10 / 10	0 / 0	44 / 44	0
world.CityMap.Kind	8 / 8	0 / 0	0 / 0	0
world.CityMap.DecorKind	15 / 15	0 / 0	0 / 0	0
world.CityMap.District	6 / 6	0 / 0	0 / 0	0
world.CityMap.Surface	6 / 6	0 / 0	0 / 0	0
world.CityMap.Obstacle	6 / 6	1 / 1	0 / 0	0
world.CityMap.Decoration	6 / 6	1 / 1	0 / 0	0
world.CityMap.CityBlock	9 / 9	1 / 1	4 / 4	0
world.CityMap.$anonymous_L72	0 / 0	0 / 0	1 / 1	0
world.CityRenderer	15 / 15	1 / 1	13 / 13	0
world.CollisionMap	0 / 0	0 / 0	1 / 1	0
world.GameWorld	27 / 27	3 / 3	34 / 34	0
lwjgl3.Lwjgl3Launcher	0 / 0	1 / 1	1 / 1	0
ScreenAdapter	0 / 0	0 / 0	7 / 7	0
Screen	0 / 0	0 / 0	7 / 7	0
Game	1 / 1	0 / 0	7 / 7	0

Clase / interfaz	A J/U	C J/U	M J/U	Dif.
ApplicationListener	0 / 0	0 / 0	6 / 6	0
utils.Disposable	0 / 0	0 / 0	1 / 1	0
audio.Music	0 / 0	0 / 0	13 / 13	0
audio.Music.OnCompletionListener	0 / 0	0 / 0	1 / 1	0

Clases propias de produccion

audio.AudioSettings <<class>>

DeadBeat/core/src/main/java/com/deadbeat/audio/AudioSettings.java:4
public final

ATRIBUTOS

-INSTANCE: AudioSettings {static, final}
-volume: float
-muted: boolean

CONSTRUCTORES EXPLICITOS

-AudioSettings()

METODOS DECLARADOS

+getInstance(): AudioSettings {static}
+getVolume(): float
+setVolume(float volume): void
+isMuted(): boolean
+setMuted(boolean muted): void
+toggleMute(): void
+getEffectiveVolume(): float

Singleton: constructor privado, INSTANCE static final y getInstance static. Las preferencias pertenecen al objeto.

audio.SoundtrackPlayer <<class>>

DeadBeat/core/src/main/java/com/deadbeat/audio/SoundtrackPlayer.java:27
public final

ATRIBUTOS

-MAX_TRACK_NUMBER: int {static, final}
-MAX_HISTORY: int {static, final}
+DEFAULT_VOLUME: float {static, final}
-random: Random {final}
-trackMetadata: Properties {final}
-history: List<Integer> {final}
-historyPosition: int
-order: int[]
-nextPosition: int
-lastPlayedIndex: int
-current: Music
-currentIndex: int
-appliedVolume: float
-audioSettings: AudioSettings {final}
-nextRequested: boolean {volatile}
-disposed: boolean

CONSTRUCTORES EXPLICITOS

+SoundtrackPlayer()
~SoundtrackPlayer(Random random)

METODOS DECLARADOS

+update(): void
+nextTrack(): void
+previousTrack(): void
-playFromHistory(int position, int direction): boolean
-playNextFresh(): void
-playIndex(int index): boolean
-pathOf(int index): String {static}

audio.SoundtrackPlayer <<class>>

DeadBeat/core/src/main/java/com/deadbeat/audio/SoundtrackPlayer.java:27
public final

-shuffleNewCycle(): void

-loadMetadata(): void

+getCurrentTitle(): String

+getCurrentArtist(): String

-metadataKey(int index, String field): String {static}

+getTrackNumber(): int

+getAvailableTrackCount(): int

+getVolume(): float

+isMuted(): boolean

+setVolume(float amount): void

+toggleMute(): void

-effectiveVolume(): float

-applyVolume(): void

-stopCurrent(): void

+dispose(): void

Implements: Disposable. appliedVolume es solo la cache del valor enviado a Music; AudioSettings conserva las preferencias.

DeadbeatGame <<class>>

DeadBeat/core/src/main/java/com/deadbeat/DeadbeatGame.java:8
public final

ATRIBUTOS

Ninguno declarado.

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

+create(): void

+dispose(): void

Extends: Game.

entities.Damageable <<interface>>

DeadBeat/core/src/main/java/com/deadbeat/entities/Damageable.java:4
public abstract

ATRIBUTOS

Ninguno declarado.

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

+takeDamage(float amount): void {abstract}

+isAlive(): boolean {abstract}

entities.ExperienceOrb <<class>>

DeadBeat/core/src/main/java/com/deadbeat/entities/ExperienceOrb.java:4
public final

ATRIBUTOS

-x: float {final}

-y: float {final}

-amount: int {final}

CONSTRUCTORES EXPLICITOS

+ExperienceOrb(float x, float y, int amount)

METODOS DECLARADOS

+getX(): float

+getY(): float

+getAmount(): int

entities.Player <<class>>

DeadBeat/core/src/main/java/com/deadbeat/entities/Player.java:23
public final

ATRIBUTOS

+DEFAULT_SPEED: float {static, final}

+DEFAULT_MAX_HEALTH: float {static, final}

+DEFAULT_RADIUS: float {static, final}

+INVULNERABILITY_SECONDS: float {static, final}

-arenaLeft: float {final}

-arenaBottom: float {final}

-arenaRight: float {final}

-arenaTop: float {final}

-spawnX: float {final}

-spawnY: float {final}

-x: float

-y: float

-radius: float {final}

-moveSpeed: float

-health: float

-maxHealth: float

-invulnerabilityRemaining: float

-facingX: float

-facingY: float

-moving: boolean

-walkTime: float

-weapons: List<Weapon> {final}

-experience: int

-level: int

CONSTRUCTORES EXPLICITOS

+Player(float centerX, float centerY, Rectangle arena)

METODOS DECLARADOS

+addExperience(int amount): int

+increaseMaxHealth(): void

+getExperience(): int

+getLevel(): int

+getExperienceToNextLevel(): int

+getWeapons(): List<Weapon>

+updateWeapons(float delta, GameWorld world): void

entities.Player <<class>>

DeadBeat/core/src/main/java/com/deadbeat/entities/Player.java:23
public final

+update(float delta): void
+update(float delta, CollisionMap solidMap): void
+update(float delta, float horizontal, float vertical): void
+update(float delta, float horizontal, float vertical, CollisionMap solidMap): void
+takeDamage(float amount): void
+heal(float amount): void
+setMoveSpeed(float newSpeed): void
+reset(): void
+isAlive(): boolean
+isInvulnerable(): boolean
+getX(): float
+getY(): float
+getRadius(): float
+getMoveSpeed(): float
+getHealth(): float
+getMaxHealth(): float
+getInvulnerabilityRemaining(): float
+getFacingX(): float
+getFacingY(): float
+isMoving(): boolean
+getWalkTime(): float
-clamp(float value, float min, float max): float {static}

Implements: Damageable.

entities.Projectile <<class>>

DeadBeat/core/src/main/java/com/deadbeat/entities/Projectile.java:3
public final

ATRIBUTOS

-x: float
-y: float
-remaining: float
-vx: float {final}
-vy: float {final}
-damage: float {final}

CONSTRUCTORES EXPLICITOS

+Projectile(float x, float y, float dx, float dy, float damage)

METODOS DECLARADOS

+update(float delta): void
+expire(): void
+isAlive(): boolean
+getX(): float
+getY(): float
+getDamage(): float
+getRadius(): float

entities.Zombie <<class>>

DeadBeat/core/src/main/java/com/deadbeat/entities/Zombie.java:11
public final

ATRIBUTOS

entities.Zombie <<class>>

DeadBeat/core/src/main/java/com/deadbeat/entities/Zombie.java:11
public final

-CHASE: MovementStrategy {static, final}
-ZIGZAG: MovementStrategy {static, final}
+COMMON_VARIANTS: int {static, final}
+TANK_VARIANTS: int {static, final}
+FAST_VARIANTS: int {static, final}
-type: ZombieType {final}
-arena: Rectangle {final}
-collisionMap: CollisionMap {final}
-visualVariant: int {final}
-x: float
-y: float
-health: float
-age: float
-facingX: float
-facingY: float
-moving: boolean
-strategy: MovementStrategy

CONSTRUCTORES EXPLICITOS

+Zombie(float x, float y, ZombieType type, Rectangle arena)
+Zombie(float x, float y, ZombieType type, Rectangle arena, CollisionMap collisionMap)

METODOS DECLARADOS

+update(float delta, Player target): void
+advance(float dx, float dy, float delta): void
+setMovementStrategy(MovementStrategy value): void
+getMovementStrategy(): MovementStrategy
+takeDamage(float damage): void
+isAlive(): boolean
+getX(): float
+getY(): float
+getRadius(): float
+getHealth(): float
+getAge(): float
+getType(): ZombieType
+getVisualVariant(): int
+getFacingX(): float
+getFacingY(): float
+isMoving(): boolean

Implements: Damageable.

entities.ZombieType <<enumeration>>

DeadBeat/core/src/main/java/com/deadbeat/entities/ZombieType.java:3
public final

ATRIBUTOS

+NORMAL: ZombieType {static, final}

+FAST: ZombieType {static, final}

+TANK: ZombieType {static, final}

+health: float {final}

+speed: float {final}

+radius: float {final}

+damage: float {final}

+experience: int {final}

CONSTRUCTORES EXPLICITOS

-ZombieType(float health, float speed, float radius, float damage, int experience)

METODOS DECLARADOS

Ninguno declarado.

Sin constructores implicitos ni values/valueOf sinteticos. Constantes enum incluidas como atributos.

graphics.FacingDirection <<enumeration>>

DeadBeat/core/src/main/java/com/deadbeat/graphics/FacingDirection.java:7
public final

ATRIBUTOS

+DOWN: FacingDirection {static, final}

+DOWN_RIGHT: FacingDirection {static, final}

+RIGHT: FacingDirection {static, final}

+UP_RIGHT: FacingDirection {static, final}

+UP: FacingDirection {static, final}

+UP_LEFT: FacingDirection {static, final}

+LEFT: FacingDirection {static, final}

+DOWN_LEFT: FacingDirection {static, final}

-DIRECTIONS: FacingDirection[] {static, final}

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

+fromVector(float dx, float dy): FacingDirection {static}

Sin constructores implicitos ni values/valueOf sinteticos. Constantes enum incluidas como atributos.

graphics.PlayerSpriteRenderer <<class>>

DeadBeat/core/src/main/java/com/deadbeat/graphics/PlayerSpriteRenderer.java:17
public final

ATRIBUTOS

-FRAME_WIDTH: int {static, final}

-FRAME_HEIGHT: int {static, final}

-STEP_SECONDS: float {static, final}

-WALK_SEQUENCE: int[] {static, final}

-texture: Texture {final}

-frames: TextureRegion[][] {final}

CONSTRUCTORES EXPLICITOS

+PlayerSpriteRenderer()

METODOS DECLARADOS

+draw(SpriteBatch batch, Player player): void

+dispose(): void

Implements: Disposable.

graphics.ZombieSpriteRenderer <<class>>

DeadBeat/core/src/main/java/com/deadbeat/graphics/ZombieSpriteRenderer.java:18
public final

ATRIBUTOS

-COMMON_FRAME_WIDTH: int {static, final}
-COMMON_FRAME_HEIGHT: int {static, final}
-TANK_FRAME_WIDTH: int {static, final}
-TANK_FRAME_HEIGHT: int {static, final}
-FAST_FRAME_WIDTH: int {static, final}
-FAST_FRAME_HEIGHT: int {static, final}
-FAST_DRAW_WIDTH: float {static, final}
-FAST_DRAW_HEIGHT: float {static, final}
-COMMON_STEP_SECONDS: float {static, final}
-TANK_STEP_SECONDS: float {static, final}
-FAST_STEP_SECONDS: float {static, final}
-COMMON_WALK_SEQUENCE: int[] {static, final}
-TANK_WALK_SEQUENCE: int[] {static, final}
-commonTextures: Texture[] {final}
-commonFrames: TextureRegion[][][] {final}
-tankTextures: Texture[] {final}
-tankFrames: TextureRegion[][][] {final}
-fastTextures: Texture[] {final}
-fastFrames: TextureRegion[][][] {final}

CONSTRUCTORES EXPLICITOS

+ZombieSpriteRenderer()

METODOS DECLARADOS

-loadCommon(): void
-loadTank(): void
-loadFast(): void
+draw(SpriteBatch batch, Zombie zombie): void
-drawCommon(SpriteBatch batch, Zombie zombie): void
-drawTank(SpriteBatch batch, Zombie zombie): void
-drawFast(SpriteBatch batch, Zombie zombie): void
~fastRowFromVector(float x, float y): int {static}
-tankRowFromVector(float x, float y): int
+dispose(): void

Implements: Disposable.

movement.ChaseMovement <<class>>

DeadBeat/core/src/main/java/com/deadbeat/movement/ChaseMovement.java:5
public final

ATRIBUTOS

Ninguno declarado.

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

+move(Zombie z, Player p, float delta): void

Implements: MovementStrategy.

movement.MovementStrategy <<interface>>

DeadBeat/core/src/main/java/com/deadbeat/movement/MovementStrategy.java:5
public abstract

ATRIBUTOS

Ninguno declarado.

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

+move(Zombie zombie, Player target, float delta): void {abstract}

movement.ZigzagMovement <<class>>

DeadBeat/core/src/main/java/com/deadbeat/movement/ZigzagMovement.java:5
public final

ATRIBUTOS

Ninguno declarado.

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

+move(Zombie z, Player p, float delta): void

Implements: MovementStrategy.

progression.Upgrade <<enumeration>>

DeadBeat/core/src/main/java/com/deadbeat/progression/Upgrade.java:8
public final

ATRIBUTOS

+GUITAR_DAMAGE: Upgrade {static, final}

+GUITAR_RATE: Upgrade {static, final}

+DRUM_DAMAGE: Upgrade {static, final}

+DRUM_RATE: Upgrade {static, final}

+MOVE_SPEED: Upgrade {static, final}

+MAX_HEALTH: Upgrade {static, final}

-label: String {final}

CONSTRUCTORES EXPLICITOS

-Upgrade(String label)

METODOS DECLARADOS

+getLabel(): String

+apply(Player player): void

Sin constructores implicitos ni values/valueOf sinteticos. Constantes enum incluidas como atributos.

progression.UpgradeSelection <<class>>

DeadBeat/core/src/main/java/com/deadbeat/progression/UpgradeSelection.java:10
public final

ATRIBUTOS

-LIMIT: int {static, final}

-counts: int[] {final}

-choices: List<Upgrade> {final}

-pending: int

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

+award(int levels): void

-offer(): void

+choose(int index, Player player): void

+getChoices(): List<Upgrade>

progression.UpgradeSelection <<class>>

DeadBeat/core/src/main/java/com/deadbeat/progression/UpgradeSelection.java:10
public final

+getPending(): int

+cancel(): void

+reset(): void

progression.Wave <<class>>

DeadBeat/core/src/main/java/com/deadbeat/progression/Wave.java:6
public final

ATRIBUTOS

-duration: float {final}

-normal: int {final}

-fast: int {final}

-tank: int {final}

CONSTRUCTORES EXPLICITOS

~Wave(float duration, int normal, int fast, int tank)

METODOS DECLARADOS

+getDuration(): float

+getTotal(): int

+typeAt(int index): ZombieType

progression.WaveBuilder <<class>>

DeadBeat/core/src/main/java/com/deadbeat/progression/WaveBuilder.java:3
public final

ATRIBUTOS

-duration: float

-normal: int

-fast: int

-tank: int

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

+duration(float value): WaveBuilder

+normal(int value): WaveBuilder

+fast(int value): WaveBuilder

+tank(int value): WaveBuilder

+build(): Wave

+campaign(float duration): Wave[] {static}

screens.GameScreen <<class>>

DeadBeat/core/src/main/java/com/deadbeat/screens/GameScreen.java:23
public final

ATRIBUTOS

+VIEW_WIDTH: float {static, final}

+VIEW_HEIGHT: float {static, final}

-GUITAR_X: float {static, final}

-GUITAR_WIDTH: float {static, final}

-GUITAR_HEIGHT: float {static, final}

-GUITAR_Y: float {static, final}

-MASK_WIDTH: int {static, final}

-MASK_HEIGHT: int {static, final}

-PLAYER_HUD_W: float {static, final}

screens.GameScreen <<class>>

DeadBeat/core/src/main/java/com/deadbeat/screens/GameScreen.java:23
public final

-PLAYER_HUD_H: float {static, final}
-PLAYER_HUD_X: float {static, final}
-PLAYER_HUD_Y: float {static, final}
-MUSIC_BUTTON_W: float {static, final}
-MUSIC_BUTTON_H: float {static, final}
-MUSIC_BUTTON_Y: float {static, final}
-PREVIOUS_X: float {static, final}
-NEXT_X: float {static, final}
-MUSIC_HUD_HOLD_SECONDS: float {static, final}
-MUSIC_HUD_FADE_SECONDS: float {static, final}
-RESULT_X: float {static, final}
-RESULT_Y: float {static, final}
-RESULT_W: float {static, final}
-RESULT_H: float {static, final}
-RESULT_BUTTON_X: float {static, final}
-RESULT_BUTTON_W: float {static, final}
-RESULT_BUTTON_H: float {static, final}
-RESULT_RETRY_Y: float {static, final}
-RESULT_EXIT_Y: float {static, final}
-MENU_X: float {static, final}
-MENU_Y: float {static, final}
-MENU_WIDTH: float {static, final}
-MENU_HEIGHT: float {static, final}
-BUTTON_X: float {static, final}
-BUTTON_WIDTH: float {static, final}
-BUTTON_HEIGHT: float {static, final}
-CONTINUE_Y: float {static, final}
-EXIT_Y: float {static, final}
-SPEAKER_X: float {static, final}
-SPEAKER_Y: float {static, final}
-SPEAKER_SIZE: float {static, final}
-VOLUME_X: float {static, final}
-VOLUME_Y: float {static, final}
-VOLUME_WIDTH: float {static, final}
-VOLUME_HEIGHT: float {static, final}
-worldViewport: FitViewport {final}
-hudViewport: FitViewport {final}
-batch: SpriteBatch {final}
-titleFont: BitmapFont {final}
-menuFont: BitmapFont {final}
-smallFont: BitmapFont {final}
-trackFont: BitmapFont {final}
-glyph: GlyphLayout {final}
-mouseInHud: Vector2 {final}
-pixel: Texture {final}
-guitarMask: Texture {final}
-guitarOutline: Texture {final}
-world: GameWorld {final}

screens.GameScreen <<class>>

DeadBeat/core/src/main/java/com/deadbeat/screens/GameScreen.java:23
public final

-soundtrack: SoundtrackPlayer {final}
-outcome: RunOutcome
-selectedResultButton: int
-runEndedAtNanos: long
-finalWave: int
-paused: boolean
-draggingVolume: boolean
-selectedButton: int
-displayedTrackNumber: int
-displayedTitle: String
-displayedArtist: String
-musicHudChangedAtNanos: long
-runStartedAtNanos: long

CONSTRUCTORES EXPLICITOS

+GameScreen()
+GameScreen(float initialVolume, boolean initiallyMuted)

METODOS DECLARADOS

+render(float delta): void
-drawExperienceBar(): void
-handleUpgradeInput(): void
-drawUpgradeSelection(): void
-checkRunOutcome(): void
-finishRun(RunOutcome result): void
-resetRun(): void
-handleResultInput(): void
-resultButtonUnderMouse(float x, float y): int
-activateResultButton(int button): void
-handleMusicHudClick(): void
-handlePauseMenuInput(): void
-changeVolumeAt(float mouseX): void
-buttonUnderMouse(float x, float y): int
-inside(float x, float y, float left, float bottom, float w, float h): boolean {static}
-activateButton(int button): void
-followPlayer(): void
-drawGuitarHealth(Player player): void
-syncMusicHudTrack(): void
-musicHudOpacity(): float
-drawMusicHud(): void
-musicBlock(float x, float y, float w, float h, float r, float g, float b, float baseAlpha, float fade): void
-drawMusicHudButton(float x, float y, String label, float fade): void
-fitting(BitmapFont font, String value, float maxWidth): String
-drawResultScreen(): void
-drawResultButton(float y, String label, boolean active, float ar, float ag, float ab): void
-drawPauseMenu(): void
-drawVolumeControl(): void
-drawPauseButton(float y, String label, boolean selected): void
-elapsedAsClock(): String

screens.GameScreen <<class>>

DeadBeat/core/src/main/java/com/deadbeat/screens/GameScreen.java:23
public final

-two(long value): String {static}

-centered(BitmapFont font, String text, float centerX, float baseline, float r, float g, float b, float a): void

-block(float x, float y, float w, float h, float r, float g, float b, float a): void

-clamp(float v, float lo, float hi): float {static}

+resize(int width, int height): void

+dispose(): void

Extends: ScreenAdapter. Los dos constructores respetan la configuracion compartida o asignan argumentos explicitamente antes de reproducir.

screens.GameScreen.RunOutcome <<enumeration>>

DeadBeat/core/src/main/java/com/deadbeat/screens/GameScreen.java:60
private static final

ATRIBUTOS

+PLAYING: RunOutcome {static, final}

+DEFEAT: RunOutcome {static, final}

+VICTORY: RunOutcome {static, final}

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

Ninguno declarado.

Tipo anidado en com.deadbeat.screens.GameScreen. Sin constructores implicitos ni values/valueOf sinteticos. Constantes enum incluidas como atributos.

screens.MainMenuScreen <<class>>

DeadBeat/core/src/main/java/com/deadbeat/screens/MainMenuScreen.java:24
public final

ATRIBUTOS

-W: float {static, final}

-H: float {static, final}

-BUTTON_X: float {static, final}

-BUTTON_W: float {static, final}

-BUTTON_H: float {static, final}

-PLAY_Y: float {static, final}

-EXIT_Y: float {static, final}

-THEME_PATH: String {static, final}

-VOLUME_X: float {static, final}

-VOLUME_Y: float {static, final}

-VOLUME_W: float {static, final}

-VOLUME_H: float {static, final}

-SPEAKER_X: float {static, final}

-SPEAKER_Y: float {static, final}

-SPEAKER_SIZE: float {static, final}

-game: DeadbeatGame {final}

-viewport: FitViewport {final}

-batch: SpriteBatch {final}

-titleFont: BitmapFont {final}

-menuFont: BitmapFont {final}

-glyph: GlyphLayout {final}

-mouse: Vector2 {final}

screens.MainMenuScreen <<class>>

DeadBeat/core/src/main/java/com/deadbeat/screens/MainMenuScreen.java:24
public final

-pixel: Texture {final}

-menuTheme: Music

-selected: int

-elapsed: float

-disposed: boolean

-draggingVolume: boolean

-audioSettings: AudioSettings {final}

-appliedThemeVolume: float

CONSTRUCTORES EXPLICITOS

+MainMenuScreen(DeadbeatGame game)

METODOS DECLARADOS

+render(float delta): void

-handleInput(): boolean

-buttonAt(float x, float y): int

-inside(float x, float y, float bx, float by, float bw, float bh): boolean {static}

-activate(int choice): boolean

-drawScenery(): void

-drawMenu(): void

-drawVolumeSlider(): void

-changeVolume(float value): void

-toggleMute(): void

-applyThemeVolume(): void

-drawButton(float y, String label, boolean active): void

-centered(BitmapFont font, String text, float cx, float baseline, float r, float g, float b, float alpha): void

-box(float x, float y, float w, float h, float r, float g, float b, float alpha): void

-stopTheme(): void

+resize(int width, int height): void

+dispose(): void

Extends: ScreenAdapter. No declara volume ni muted independientes; usa AudioSettings y una cache de aplicacion.

weapons.Drumsticks <<class>>

DeadBeat/core/src/main/java/com/deadbeat/weapons/Drumsticks.java:6
public final

ATRIBUTOS

Ninguno declarado.

CONSTRUCTORES EXPLICITOS

+Drumsticks()

METODOS DECLARADOS

#performAttack(Player p, GameWorld world): boolean

Extends: Weapon.

weapons.ElectricGuitar <<class>>

DeadBeat/core/src/main/java/com/deadbeat/weapons/ElectricGuitar.java:6
public final

ATRIBUTOS

Ninguno declarado.

CONSTRUCTORES EXPLICITOS

+ElectricGuitar()

METODOS DECLARADOS

#performAttack(Player p, GameWorld world): boolean

Extends: Weapon.

weapons.Weapon <<class>>

DeadBeat/core/src/main/java/com/deadbeat/weapons/Weapon.java:7
public abstract

ATRIBUTOS

-damage: float

-interval: float

-remaining: float

CONSTRUCTORES EXPLICITOS

#Weapon(float damage, float interval)

METODOS DECLARADOS

+update(float delta, Player player, GameWorld world): void {final}

#performAttack(Player player, GameWorld world): boolean {abstract}

+getDamage(): float

+getInterval(): float

+improveDamage(): void

+improveRate(): void

world.CityMap <<class>>

DeadBeat/core/src/main/java/com/deadbeat/world/CityMap.java:15
public final

ATRIBUTOS

+WORLD_SIZE: float {static, final}

+SPAWN: float {static, final}

+TILE_SIZE: int {static, final}

+BLOCK_SIZE: int {static, final}

+STREET_X: int {static, final}

+STREET_Y: int {static, final}

-CACHE_LIMIT: int {static, final}

-SEED: int {static, final}

-brokenGarbageIds: Set<Long> {final}

-cache: LinkedHashMap<Long, CityBlock> {final}

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

+variation(int x, int y, int salt): int {static}

-positive(int hash, int modulo): int {static}

+avenueCenter(int i): float

+streetCenter(int j): float

+avenueHalf(int i): float

+streetHalf(int j): float

+avenueIndex(float x): int

+streetIndex(float y): int

world.CityMap <<class>>

DeadBeat/core/src/main/java/com/deadbeat/world/CityMap.java:15
public final

-nearestAvenue(float x): float

-nearestStreet(float y): float

+isRoad(float x, float y): boolean

+surfaceAt(float x, float y): Surface

+isAvenue(float x): boolean

+isStreet(float y): boolean

+nearestAvenueCenter(float x): float

+nearestStreetCenter(float y): float

+getBlock(int i, int j): CityBlock

+cachedBlockCount(): int

+districtAt(float x, float y): District

-generateBlock(int bx, int by): CityBlock

-resolveLayout(List<Obstacle> solids, List<Decoration> decals, float west, float south, float east, float north): void

-resolveObstacleOverlaps(List<Obstacle> solids, float west, float south, float east, float north): void

-relocateObstacle(Obstacle o, List<Obstacle> solids, float west, float south, float east, float north): Obstacle

-relocateDecoration(Decoration d, List<Decoration> placed, List<Obstacle> solids, float west, float south, float east, float north): Decoration

-avoidDecorationOverlap(DecorKind kind): boolean {static}

-obstaclePadding(Kind kind): float {static}

-decorationPadding(DecorKind kind): float {static}

-overlapsAny(Obstacle o, List<Obstacle> others, float padding): boolean {static}

-overlapsAny(Decoration d, List<Obstacle> others, float padding): boolean {static}

-overlapsAnyDecor(Decoration d, List<Decoration> others, float padding): boolean {static}

-intersects(float ax, float ay, float aw, float ah, float bx, float by, float bw, float bh, float padding): boolean {static}

-clamp(float value, float min, float max): float {static}

-addBuilding(List<Obstacle> solids, float x, float y, float w, float h, int style): void {static}

-add(List<Obstacle> solids, float x, float y, float w, float h, Kind kind, int style): void {static}

-addDec(List<Decoration> list, float x, float y, float w, float h, DecorKind kind, int style): void {static}

-garbageId(int bx, int by, int decorationIndex): long {static}

+breakGarbageNear(float x, float y, float playerRadius): int

+brokenGarbageVariant(int bx, int by, int decorationIndex): int

+isGarbageBroken(int bx, int by, int decorationIndex): boolean

+brokenGarbageCount(): int

+resetGarbage(): void

-circleIntersectsRect(float x, float y, float radius, float left, float bottom, float width, float height): boolean {static}

-solidDecoration(DecorKind kind): boolean {static}

+collidesCircle(float x, float y, float radius): boolean

Implements: CollisionMap.

world.CityMap.Kind <<enumeration>>

DeadBeat/core/src/main/java/com/deadbeat/world/CityMap.java:25
public static final

ATRIBUTOS

+BUILDING: Kind {static, final}

+CAR: Kind {static, final}

+DUMPSTER: Kind {static, final}

+BARRICADE: Kind {static, final}

+PLANTER: Kind {static, final}

+FENCE: Kind {static, final}

+TENT: Kind {static, final}

+CRATE: Kind {static, final}

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

Ninguno declarado.

Tipo anidado en com.deadbeat.world.CityMap. Sin constructores implicitos ni values/valueOf sinteticos. Constantes enum incluidas como atributos.

world.CityMap.DecorKind <<enumeration>>

DeadBeat/core/src/main/java/com/deadbeat/world/CityMap.java:26
public static final

ATRIBUTOS

+TRASH: DecorKind {static, final}

+PAPER: DecorKind {static, final}

+GRAFFITI: DecorKind {static, final}

+BENCH: DecorKind {static, final}

+LAMP: DecorKind {static, final}

+WEEDS: DecorKind {static, final}

+PUDDLE: DecorKind {static, final}

+MANHOLE: DecorKind {static, final}

+POWER: DecorKind {static, final}

+BARREL: DecorKind {static, final}

+VENDING: DecorKind {static, final}

+TIRES: DecorKind {static, final}

+PALLET: DecorKind {static, final}

+BIN: DecorKind {static, final}

+FENCE: DecorKind {static, final}

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

Ninguno declarado.

Tipo anidado en com.deadbeat.world.CityMap. Sin constructores implicitos ni values/valueOf sinteticos. Constantes enum incluidas como atributos.

world.CityMap.District <<enumeration>>

DeadBeat/core/src/main/java/com/deadbeat/world/CityMap.java:27
public static final

ATRIBUTOS

+SHOPS: District {static, final}

+RESIDENTIAL: District {static, final}

+INDUSTRIAL: District {static, final}

+PARKING: District {static, final}

+CAMP: District {static, final}

+BACKSTREET: District {static, final}

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

Ninguno declarado.

Tipo anidado en com.deadbeat.world.CityMap. Sin constructores implicitos ni values/valueOf sinteticos. Constantes enum incluidas como atributos.

world.CityMap.Surface <<enumeration>>

DeadBeat/core/src/main/java/com/deadbeat/world/CityMap.java:28
public static final

ATRIBUTOS

+ROAD: Surface {static, final}

+SIDEWALK: Surface {static, final}

+BRICK: Surface {static, final}

+ALLEY: Surface {static, final}

+LOT: Surface {static, final}

+DIRT: Surface {static, final}

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

Ninguno declarado.

Tipo anidado en com.deadbeat.world.CityMap. Sin constructores implicitos ni values/valueOf sinteticos. Constantes enum incluidas como atributos.

world.CityMap.Obstacle <<class>>

DeadBeat/core/src/main/java/com/deadbeat/world/CityMap.java:30
public static final

ATRIBUTOS

+x: float {final}

+y: float {final}

+width: float {final}

+height: float {final}

+kind: Kind {final}

+style: int {final}

CONSTRUCTORES EXPLICITOS

+Obstacle(float x, float y, float width, float height, Kind kind, int style)

METODOS DECLARADOS

Ninguno declarado.

Tipo anidado en com.deadbeat.world.CityMap.

world.CityMap.Decoration <<class>>

DeadBeat/core/src/main/java/com/deadbeat/world/CityMap.java:40
public static final

ATRIBUTOS

```
+x: float {final}
+y: float {final}
+width: float {final}
+height: float {final}
+kind: DecorKind {final}
+style: int {final}
```

CONSTRUCTORES EXPLICITOS

```
+Decoration(float x, float y, float width, float height, DecorKind kind, int style)
```

METODOS DECLARADOS

Ninguno declarado.

Tipo anidado en com.deadbeat.world.CityMap.

world.CityMap.CityBlock <<class>>

DeadBeat/core/src/main/java/com/deadbeat/world/CityMap.java:50
public static final

ATRIBUTOS

```
+blockX: int {final}
+blockY: int {final}
+district: District {final}
+left: float {final}
+bottom: float {final}
+right: float {final}
+top: float {final}
-obstacles: Obstacle[] {final}
-decorations: Decoration[] {final}
```

CONSTRUCTORES EXPLICITOS

```
~CityBlock(int x, int y, District district, float left, float bottom, float right, float top,
Obstacle[] obstacles, Decoration[] decorations)
```

METODOS DECLARADOS

```
+obstacleCount(): int
+obstacle(int i): Obstacle
+decorationCount(): int
+decoration(int i): Decoration
```

Tipo anidado en com.deadbeat.world.CityMap.

world.CityMap.cache (clase anonima) <<anonymous>>

DeadBeat/core/src/main/java/com/deadbeat/world/CityMap.java:72

ATRIBUTOS

Ninguno declarado.

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

```
#removeEldestEntry(Map.Entry<Long, CityBlock> eldest): boolean
```

Tipo anidado en com.deadbeat.world.CityMap. Extends: LinkedHashMap<Long, CityBlock>. La etiqueta describe el objeto anonimo del cache; no inventa un nombre de clase Java.

world.CityRenderer <<class>>

DeadBeat/core/src/main/java/com/deadbeat/world/CityRenderer.java:22
public final

ATRIBUTOS

world.CityRenderer <<class>>

DeadBeat/core/src/main/java/com/deadbeat/world/CityRenderer.java:22
public final

-TILE: int {static, final}
-map: CityMap {final}
-SURFACE_NAMES: String[] {static, final}
-ASPHALT: int {static, final}
-SIDEWALK: int {static, final}
-BRICK: int {static, final}
-ALLEY: int {static, final}
-LOT: int {static, final}
-DIRT: int {static, final}
-ROOF: int {static, final}
-surfaceTextures: Texture[] {final}
-surfaceRegions: TextureRegion[] {final}
-sprites: Map<String, Texture> {final}
-pixel: Texture {final}
-FILES: String[] {static, final}

CONSTRUCTORES EXPLICITOS

+CityRenderer(CityMap map, Texture pixel)

METODOS DECLARADOS

+draw(SpriteBatch batch, float left, float bottom, float w, float h): void
-drawGround(SpriteBatch batch, float left, float bottom, float w, float h): void
-drawMaterial(SpriteBatch batch, int material, float x, float y, float width, float height): void
~worldTileOrigin(int worldPixel): int {static}
-drawClippedMaterial(SpriteBatch batch, int material, float x, float y, float width, float height, float viewLeft, float viewBottom, float viewWidth, float viewHeight): void
-drawRoadEdges(SpriteBatch batch, float left, float bottom, float w, float h): void
-drawObstacle(SpriteBatch b, Obstacle o): void
-drawBuilding(SpriteBatch b, Obstacle o): void
-drawDecoration(SpriteBatch b, Decoration d, boolean broken, int variant): void
-sprite(SpriteBatch batch, String key, float x, float y, float w, float h): void
-rect(SpriteBatch b, float x, float y, float w, float h, float r, float g, float bl): void
-visible(float x, float y, float w, float h, float left, float bottom, float vw, float vh): boolean {static}
+dispose(): void

Implements: Disposable.

world.CollisionMap <<interface>>

DeadBeat/core/src/main/java/com/deadbeat/world/CollisionMap.java:4
public abstract

ATRIBUTOS

Ninguno declarado.

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

+collidesCircle(float centerX, float centerY, float radius): boolean {abstract}

world.GameWorld <<class>>

DeadBeat/core/src/main/java/com/deadbeat/world/GameWorld.java:28
public final

ATRIBUTOS

-MAX_ZOMBIES: int {static, final}

world.GameWorld <<class>>

DeadBeat/core/src/main/java/com/deadbeat/world/GameWorld.java:28
public final

-DRUMSTICKS_EFFECT_DURATION: float {static, final}
-DRUMSTICKS_EFFECT_SIZE: float {static, final}
-NOTE_PROJECTILE_SIZE: float {static, final}
-upgrades: UpgradeSelection {final}
-waves: Wave[] {final}
-random: Random {final}
-waveIndex: int
-spawned: int
-waveTime: float
-wavesFinished: boolean
-pixel: Texture {final}
-cityMap: CityMap {final}
-cityRenderer: CityRenderer {final}
-player: Player {final}
-playerSpriteRenderer: PlayerSpriteRenderer {final}
-zombieSpriteRenderer: ZombieSpriteRenderer {final}
-drumsticksAttackTexture: Texture {final}
-noteProjectileTexture: Texture {final}
-drumsticksAttackFrames: TextureRegion[] {final}
-noteProjectileFrames: TextureRegion[] {final}
-arena: Rectangle {final}
-zombies: List<Zombie> {final}
-projectiles: List<Projectile> {final}
-orbs: List<ExperienceOrb> {final}
-elapsedTime: float
-pulse: float

CONSTRUCTORES EXPLICITOS

+GameWorld()
+GameWorld(boolean graphics)
+GameWorld(boolean graphics, float waveDuration)

METODOS DECLARADOS

+update(float delta): void
+update(float delta, float horizontal, float vertical): void
-updateCombat(float delta): void
-collectExperience(): void
+isChoosingUpgrade(): boolean
+getUpgradeChoices(): List<Upgrade>
+getPendingUpgrades(): int
+chooseUpgrade(int index): void
+cancelUpgradeSelection(): void
+getOrbs(): List<ExperienceOrb>
-updateSpawning(float delta): void
-advanceWave(): void
-spawn(ZombieType type): boolean
+getCurrentWave(): int
+getWaveRemaining(): float
+areWavesFinished(): boolean
-damage(Damageable target, float amount): void {static}

world.GameWorld <<class>>

DeadBeat/core/src/main/java/com/deadbeat/world/GameWorld.java:28
public final

+touch(float x, float y, float radius, float otherX, float otherY, float otherRadius): boolean {static}

+findNearestZombie(float x, float y, float range): Zombie

+addProjectile(Projectile projectile): void

+strikeNearby(float range, float amount): boolean

+getZombies(): List<Zombie>

+getProjectileCount(): int

+reset(): void

+draw(SpriteBatch batch, float viewLeft, float viewBottom, float viewWidth, float viewHeight): void

-drawGuitarProjectile(SpriteBatch batch, Projectile projectile): void

-drawDrumsticksSpin(SpriteBatch batch): void

-drawPlayer(SpriteBatch batch): void

-rect(SpriteBatch batch, float x, float y, float w, float h, float red, float green, float blue): void

-splitHorizontalStrip(Texture texture): TextureRegion[] {static}

+getPlayer(): Player

+getCityMap(): CityMap

+getElapsedTime(): float

+dispose(): void

Implements: Disposable. Relaciones de renderizado: GameWorld crea y libera pixel, drumsticksAttackTexture y noteProjectileTexture (Texture). drumsticksAttackFrames y noteProjectileFrames contienen referencias TextureRegion obtenidas por splitHorizontalStrip; las regiones no poseen el ciclo de vida de las texturas.

lwjgl3.Lwjgl3Launcher <<class>>

DeadBeat/lwjgl3/src/main/java/com/deadbeat/lwjgl3/Lwjgl3Launcher.java:8
public final

ATRIBUTOS

Ninguno declarado.

CONSTRUCTORES EXPLICITOS

-Lwjgl3Launcher()

METODOS DECLARADOS

+main(String[] args): void {static}

Bases e interfaces de apoyo de libGDX

ScreenAdapter <<class>>

libGDX 1.14.2 sources / ScreenAdapter.java
public

ATRIBUTOS

Ninguno declarado.

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

+render(float delta): void

+resize(int width, int height): void

+show(): void

+hide(): void

+pause(): void

+resume(): void

+dispose(): void

Screen <<interface>>

libGDX 1.14.2 sources / Screen.java
abstract public

ATRIBUTOS

Ninguno declarado.

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

+show(): void {abstract}

+render(float delta): void {abstract}

+resize(int width, int height): void {abstract}

+pause(): void {abstract}

+resume(): void {abstract}

+hide(): void {abstract}

+dispose(): void {abstract}

Game <<class>>

libGDX 1.14.2 sources / Game.java
public abstract

ATRIBUTOS

#screen: Screen

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

+dispose(): void

+pause(): void

+resume(): void

+render(): void

+resize(int width, int height): void

+setScreen(Screen screen): void

+getScreen(): Screen

ApplicationListener <<interface>>

libGDX 1.14.2 sources / ApplicationListener.java
abstract public

ATRIBUTOS

Ninguno declarado.

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

+create(): void {abstract}

+resize(int width, int height): void {abstract}

+render(): void {abstract}

+pause(): void {abstract}

+resume(): void {abstract}

+dispose(): void {abstract}

utils.Disposable <<interface>>

libGDX 1.14.2 sources / Disposable.java
abstract public

ATRIBUTOS

Ninguno declarado.

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

+dispose(): void {abstract}

audio.Music <<interface>>

libGDX 1.14.2 sources / Music.java
abstract public

ATRIBUTOS

Ninguno declarado.

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

+play(): void {abstract}

+pause(): void {abstract}

+stop(): void {abstract}

+isPlaying(): boolean {abstract}

+setLooping(boolean isLooping): void {abstract}

+isLooping(): boolean {abstract}

+setVolume(float volume): void {abstract}

+getVolume(): float {abstract}

+setPan(float pan, float volume): void {abstract}

+setPosition(float position): void {abstract}

+getPosition(): float {abstract}

+dispose(): void {abstract}

+setOnCompletionListener(OnCompletionListener listener): void {abstract}

audio.Music.OnCompletionListener <<interface>>

libGDX 1.14.2 sources / OnCompletionListener.java
abstract public static

ATRIBUTOS

Ninguno declarado.

CONSTRUCTORES EXPLICITOS

Ninguno declarado.

METODOS DECLARADOS

+onCompletion(Music music): void {abstract}

Tipo anidado en com.badlogic.gdx.audio.Music.